

⇒ **CONVERSIONS:-** where B-binary, D-Decimal, O-Octal, H-Hexadecimal
 $\Delta p = \text{power}$

* $B \rightarrow D \Rightarrow \times 2^p$

$B \rightarrow O \Rightarrow \text{grp of 3 bits}$

$B \rightarrow H \Rightarrow \text{grp of 4 bits}$

* $D \rightarrow B \Rightarrow \% 2$

$D \rightarrow O \Rightarrow \% 8$

$D \rightarrow H \Rightarrow \% 16$

* $O \rightarrow B \Rightarrow \text{grp of 3 bits}$

$O \rightarrow D \Rightarrow \times 8^p$

$O \rightarrow H \Rightarrow O \rightarrow B \rightarrow 4 \text{ bits grp (H)}$

* $H \rightarrow B \Rightarrow \text{grp of 4 bits}$

$H \rightarrow D \Rightarrow \times 16^p$

$H \rightarrow O \Rightarrow H \rightarrow B \rightarrow 3 \text{ bits grp (O)}$

⇒ **IDENTITIES:**

1. $A+0 = A$

2. $A \cdot 1 = A$

3. $A+1 = 1$

4. $A \cdot 0 = 0$

5. $A+A = A$

6. $A \cdot A = A$

7. $A+\bar{A} = 1$

8. $A \cdot \bar{A} = 0$

9. $A \cdot (B+C) = AB+AC$

10. $A+(BC) = (A+B)(A+C)$

11. $A+AB = A$

12. $A(A+B) = A$

13. $A+\bar{A}B = (A+B)$

14. $A(\bar{A}+B) = AB$

15. $AB+A\bar{B} = A$

16. $(A+B)(A+\bar{B}) = A$

17. $AB+\bar{A}C = (A+C)(\bar{A}+B)$

18. $(A+B)(A+C) = AC+\bar{A}B$

19. $AB+\bar{A}C+BC = AB+\bar{A}C$

20. $(A+B)(\bar{A}+C)(B+C) = (A+B)(\bar{A}+C)$

21. $\overline{A \cdot B \cdot C \dots} = \bar{A} + \bar{B} + \bar{C} + \dots$

22. $\overline{A+B+C+\dots} = \bar{A} \cdot \bar{B} \cdot \bar{C} \cdot \dots$

⇒ **CONVERSIONS:**

1. Binary to Gray Code = (1-15) zig-zag $00=0, 11=0, 01/10=1$

2. Binary to Excess-3 = (1-15... & onwards) addition of 3 to binary

3. Binary Coded Decimal (BCD) to XS-3 = addition of 3 to binary (0-9)

⇒ **BINARY ADDITION RULES:** $0+0=0$ $1+1=10$

$0+1=1$ $1+1+1=11$

$1+0=1$

BINARY SUBTRACTION RULES:

$0-0=0$ $1-0=1$

$1-1=0$ $0-1=1$ (-1 can't be taken in binary).

Teacher's Sign. :

Expression	Symbol	Venn diagram	Boolean Algebra	Truth Table Values		
				A	B	Outputs=Y
AND			$Y = A \cdot B$	0	0	0
				0	1	0
				1	0	0
				1	1	1
OR			$Y = A + B$	0	0	0
				0	1	1
				1	0	1
				1	1	1
XOR			$Y = \bar{A} \cdot B + A \cdot \bar{B}$ $Y = A \oplus B$	0	0	0
				0	1	1
				1	0	1
				1	1	0
NAND			$Y = \overline{A \cdot B}$	0	0	1
				0	1	1
				1	0	1
				1	1	0
NOR			$Y = \overline{A + B}$	0	0	1
				0	1	0
				1	0	0
				1	1	0
XNOR			$Y = \bar{A} \cdot B + A \cdot \bar{B}$ $Y = \overline{A \oplus B}$ $Y = A \odot B$	0	0	1
				0	1	0
				1	0	0
				1	1	1
NOT			$Y = \bar{A}$	0		1
				1		0

EXPERIMENT :

No.

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→ **BOOLEAN ALGEBRA PROPERTIES**

1. Identity - $A \cdot 1 = A$ $A + 0 = A$
2. Commutative - $A + B = B + A$ $A \cdot B = B \cdot A$
3. Distributive - $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$ $A + (B \cdot C) = (A + B) \cdot (A + C)$
4. Associative - $A + (B + C) = (A + B) + C$ $A \cdot (B \cdot C) = (A \cdot B) \cdot C$
5. Complement - $A \cdot A' = 0$ $A + A' = 1$
6. Demopotency - $A + A = 1$ $A \cdot A = A$
7. Absorption - $A + 1 = 1 \Rightarrow A + AB = A(1 + AB) = A \Rightarrow A(A + B) = AA + AB = A$
8. Involution - $A'' = A$
9. Demorgan's - $(A + B)' = A' \cdot B'$ $(A \cdot B)' = A' + B'$
10. std. POS - add $[C \cdot \bar{C}]$
11. std. SOP - add $[C + \bar{C}]$

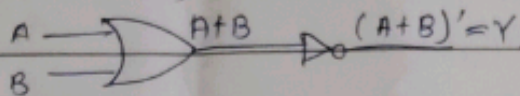
De Morgan's Theorem's Proof :-

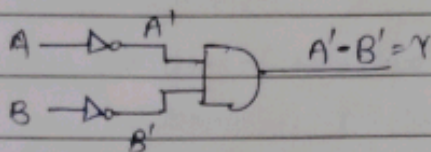
i) $(A+B)' = A' \cdot B'$ - 1st

ii) $(A \cdot B)' = A' + B'$ - 2nd

Proof :- The complement of a sum is equal to the complement $(A+B)' = A' \cdot B'$

De Morgan's 1st Law :- $(A+B)' = A' \cdot B'$

Logical diagram :-  $(A+B)' = Y$

T.T

Similarly for De Morgan's 2nd Law :- $(A \cdot B)' = A' + B'$

17-1-2026

Universal Gates :- 1) are capable of implementing any boolean fⁿ without requiring any other type of gate.

2) can realise all binary operations & the basic logic gates can be derived from them.

3) NAND & NOR gates are known as universal gates any logic fⁿ & basic gates can be implemented using NAND & NOR gates.

properties :- They are not associative in nature.

They are commutative in nature.

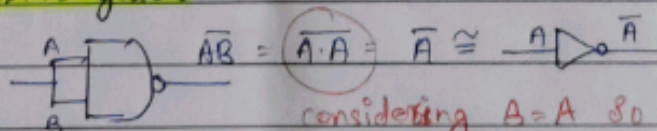
Implementation of Other gates using Universal gates :-

i) NOT Gate using NAND gate :-

$$\text{NOT} = \bar{A}$$

$$\text{AND} = A \cdot B$$

$$\text{NAND} = \overline{A \cdot B}$$



considering $A=A$ so we get : $\overline{A \cdot A} = \bar{A}$

So, if taking both inputs together we get $A=B$

$$\therefore \overline{A \cdot A} = \bar{A}$$

2) AND gate using NAND gate:

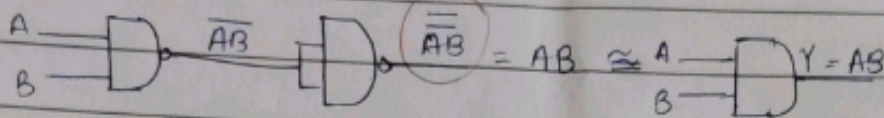
We know $AND = A \cdot B$ or $\overline{\overline{A \cdot B}} = AB$

So, $NAND = \overline{AB}$

We know,

$$\overline{\overline{A}} = A$$

$$\overline{\overline{\overline{A}}} = \overline{A}$$

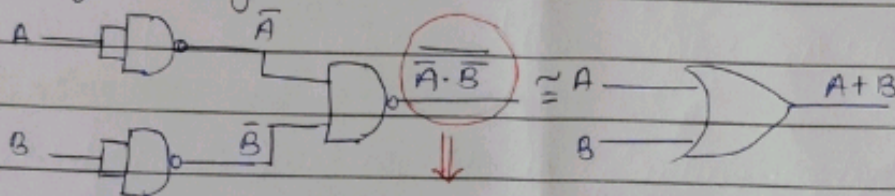


3) OR gate using NAND gate:

We know $OR = A + B$

double inversion of $A+B$ we get $\overline{\overline{A+B}} = A+B$

applying De-Morgan's 1st theorem $\overline{\overline{A+B}} = \overline{\overline{A} \cdot \overline{B}}$



explanation of dia $\Rightarrow \overline{\overline{A+B}} = \overline{\overline{A} \cdot \overline{B}}$ - De Morgan's 1st Law

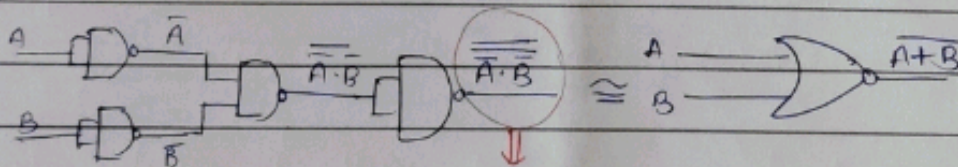
$$\overline{\overline{A} \cdot \overline{B}} = \overline{\overline{\overline{A+B}}}$$

$$\overline{\overline{\overline{A+B}}} = A+B //$$

4) NOR gate using NAND gate:

We know $NOR = \overline{A+B}$

$NAND = \overline{AB}$



logical derivation:- $\overline{\overline{A} \cdot \overline{B}} = \overline{A+B}$ 1st demorgan's law. $\left\{ \begin{array}{l} \overline{\overline{A}} = A \\ \overline{\overline{B}} = B \\ \overline{\overline{\overline{A+B}}} = \overline{A+B} \end{array} \right.$

~~A+B~~

~~A+B~~

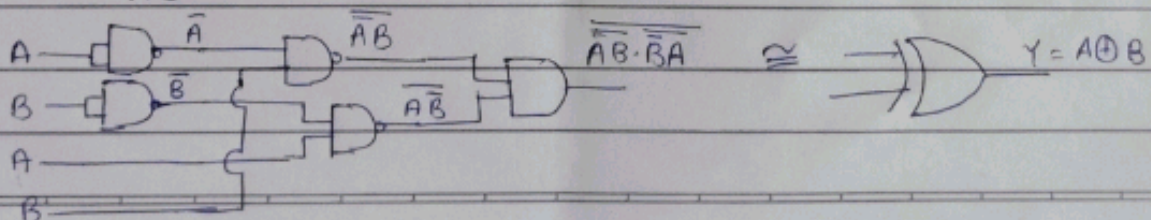
~~A+B~~

$NOR = \overline{A+B}$

5) EX-OR Gate using NAND gate:

EX-OR / XOR $= \overline{A}B + A\overline{B}$ $Y = A \oplus B$

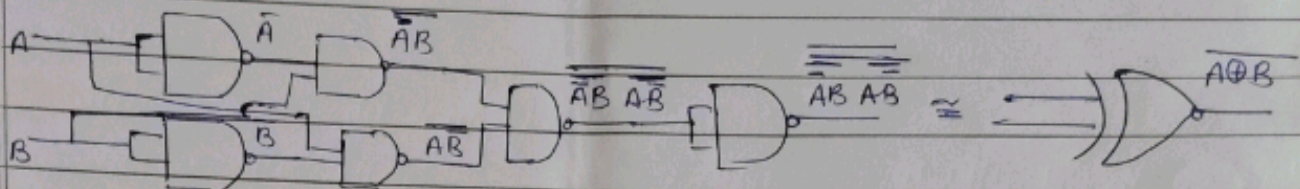
$NAND = \overline{AB}$



6. EX-NOR gate using NAND gate:

$$\text{EX-NOR} = \overline{A \oplus B}$$

$$\text{NAND} = \overline{AB}$$

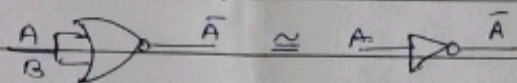


Implementation of other gates using NOR gate:

1. NOT using NOR

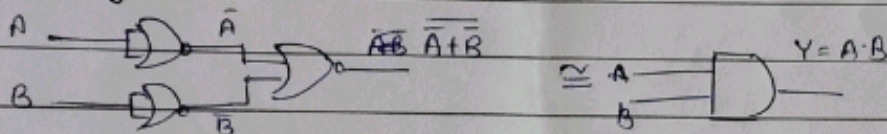
$$\text{NOR} = \overline{A+B} ; \text{NOT} = \overline{A}$$

$$\text{So, } A + \overline{A} = \overline{A}$$



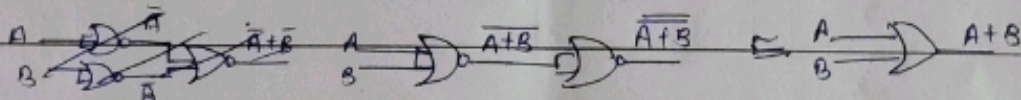
2. AND using NOR

$$\text{NOR} = \overline{A+B} \quad \text{AND} = AB$$



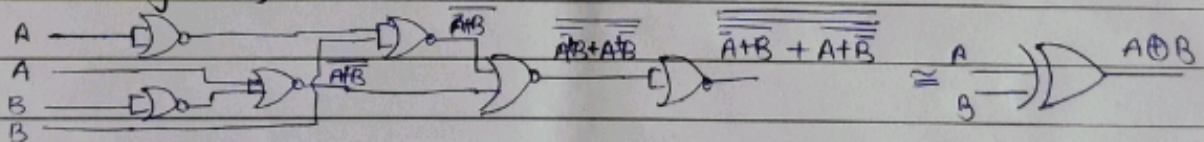
3. OR using NOR

$$\text{NOR} = \overline{A+B} ; \text{OR} = A+B$$

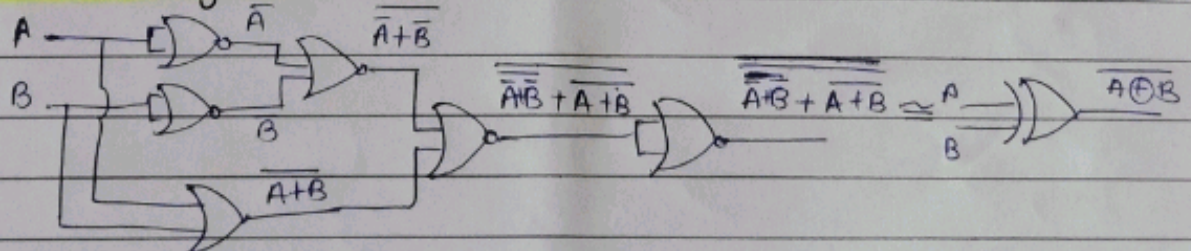


4. EX-OR using NOR

$$\text{EX-OR} = A \oplus B \quad \text{NOR} = \overline{A+B}$$



5. EX-NOR using NOR



1. **SOP** - Sum of product (1) $ABC + \bar{A}\bar{B}\bar{C}$ Σm_i (minterms)
 2. **POS** - Product of sum (0) $(A+B+C)(\bar{A}+\bar{B}+\bar{C})$ ΠM_i (maxterms)

'i' is the decimal no. equivalent of the natural binary no. with these shorthand notations logical 'f'

3. **K-MAP** - 2, 3 & 4 variables (SOP entry 1) & (POS entry 0)

A \ B	0	1
0	0	1
1	2	3

2-variable

A \ BC	00	01	11	10
0	0	1	3	2
1	4	5	7	6

3-variable

AB \ CD	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

4-variable

- **grouping of variables in K-map** :-

conditions :- i) group must no. of 1 as it follows

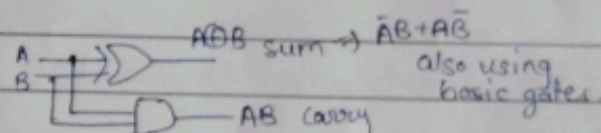
binary rule of 1, 2, 4, 8, 16, 32.

ii) pair - group of 2 adjacent 1

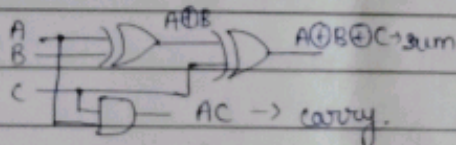
quad - " " 4 " "

octet - " " 8 " "

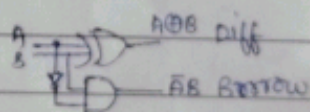
4. **HALF ADDER** :- $[(0,0), (1,1)] = 0$ (sum)
 $[1,1] = 1$ (carry)



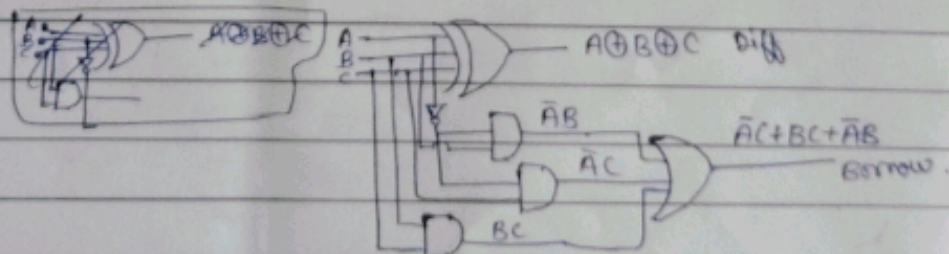
5. **FULL ADDER** :-



6. **SUBTRACTOR** :- $[(0,0), (1,0)] = 0$ & $[(1,0), (0,1)] = 1$ Difference.
 $[(0,0), (1,1), (1,0)] = 0$ & $[(0,1)] = 1$ Borrow.



7. **FULL SUBTRACTOR** :-



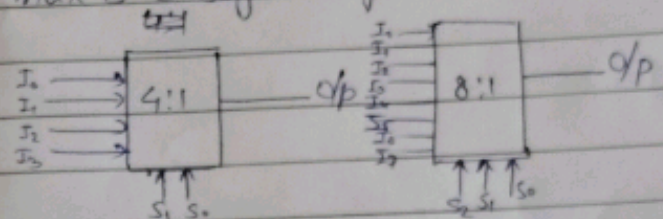
8. **MULTIPLEXER** (i) multiple inputs one output.

(ii) Its f^n is to ^{select &} connect one I/p line & connect it to O/p line, known as 'Data Selector'.

(iii) there are 2^n I/p lines where n = no. of select lines.

(iv) Depending upon no. of selecting lines mux is classified as follows:

	select line	I/p	O/p
2:1	1 ($2^1 = 2$)	2	1
4:1	2 ($2^2 = 4$)	4	1
8:1	3 ($2^3 = 8$)	8	1
16:1	4 ($2^4 = 16$)	16	1

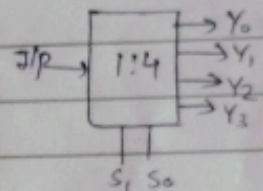


9. **DEMUX** : i) One input many outputs.

I/P	O/P	SELECT	DEMUX
1	2	1 (2^1)	1:2
1	4	2 (2^2)	1:4
1	8	3 (2^3)	1:8
1	16	4 (2^4)	1:16

Its application

- Comp memory
- communication sys
- Telephone network
- Data Transmission.

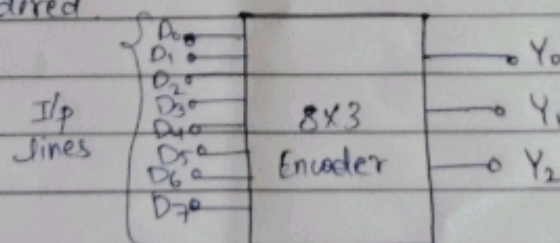


v) ALU & serial to parallel conversion

10. **ENCODER** : ^{operation} combinational circuit designed to perform inverse of decoder.

It has n no. of I/p lines & m no. of O/p lines. It produces m bit binary codes corresponds to its I/p. ex:- Priority encoder, Dec to BCD, Oct to Bin, Hex to Bin.

11. **PRIORITY ENCODER** : Special type of encoder priority given to the input lines if 2 or more input lines are 1 at the same time then I/p lines with highest priority will be considered.



SEQUENTIAL CIRCUIT: A circuit in which the I/O depends on both the present & past state of input is called as sequential circuit.

SYNCHRONOUS

1. In synchronous circuit memory elements are flip-flop.

2. The change in input signals can affect memory elements upon activation of clock signal.

3. The maximum operating speed of clock depends on time delay's in watt.

4. Easy to Design

ASYNCHRONOUS.

1. Asynchronous circuits memory elements are either clock-flip-flop time delay element.

2. change in input signals can affect memory element at any instant of time.

3. Bez of absence of clock asyn. ckts can operate far faster than syn ckts.

4. Difficult to design.

D-flip-flop

1. delay type
2. Available commercially
3. ^{can be} constructed from SR & JK
4. J & K I/p's are connected together through NOT gate.
5. used as delay device

T-flip-flop.

- toggle type
- not commercial avail.
- cannot be.
- J & K I/p's are directly connected to each other
- can be used as delay device.